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[23CSE109]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2024

COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	List the two types of DNS message	L-1	CO1	**02**
	b.	What is a layered Network Architecture?	L-1	CO2	**02**
	c.	Justify the need for layer five in the OSI model.	L-2	CO3	**02**
	d.	Discuss dynamic IP addressing.	L-2	CO4	**02**
	e.	State the purpose of User Datagram Protocol (UDP).	L-1	CO4	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Analyze the layers of the internet model and correlate to the layers of the OSI model.	L-2	CO1	**05**
		OR			
	b.	Discuss the File Transfer Protocol in detail.	L-2	CO2	**05**
3.	**a.**	Analyze why HTTP is defined as a stateless protocol.	L-2	CO4	**05**
		OR			
	b.	Identify and explain the functions of Application Layer.	L-2	CO2	**05**
4.	**a.**	Differentiate between TCP and UDP.	L-2	CO2	**05**
		OR			
	b.	List and explain the various congestion control techniques in TCP.	L-2	CO2	**05**
5.	**a.**	Suppose a TCP connection is transferring a file of 5000 bytes. The first byte is numbered 10001.What are the sequence numbers for each segment if data are sent in three segments, each carrying BTL 6 creating 1000 bytes.	L-3	CO2	**05**
		OR			
	b.	Compare and contrast OSI and TCP/IP models.	L-2	CO3	**05**
6.	**a.**	Discuss in detail about the functions of network layer and transport layer with necessary diagrams.	L-2	CO3	**05**
		OR			
	b.	Explain Link State Routing Protocol with suitable diagram.	L-2	CO2	**05**

7.	**a.**	Identify the limitations of current Internet Protocol version 4 (IPv4) and explain the features of Internet Protocol version 6 (IPv6).	L-2	CO4	**05**
		OR			
	b.	Describe the seven states of OSPF in brief.	L-2	CO2	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Illustrate the working of CSMA / CD and CSMA/CA protocol.	L-2	CO2	**10**
		OR			
	b.	Explain in detail about the Point-to-Point Protocol (PPP) with neat sketch.	L-2	CO2	**10**
9.	**a.**	Explain VLAN Trunk Protocol – VTP in detail.	L-2	CO4	**10**
		OR			
	b.	Describe the working of Network Address Translation.	L-2	CO4	**10**
10.	**a.**	Explain the role of NAT in the network layer.	L-2	CO4	**10**
		OR			
	b.	Evaluate advantages and disadvantages of slotted aloha.	L-2	CO4	**10**
11.	**a.**	Illustrate the application of distance vector routing.	L2	CO3	**10**
		OR			
	b.	Explain EIGRP in detail.	L2	CO3	**10**
12.	**a.**	Compare the various wireless technologies. Enumerate 802.11 protocol stacks in detail.	L-3	CO5	**10**
		OR			
	b.	Discuss the basic operation of DHCP.	L-2	CO5	**10**
13.	**a.**	Explain the role of 802.11.	L-2	CO6	**10**
		OR			
	b.	List out four different types of DHCP messages and explain	L-2	CO6	**10**